



DATA SCIENCE & ANALYTICS · CLIENT ENGAGEMENT BRIEF

Project FORESIGHT

Demand & Inventory Intelligence

Client: NorthBay Living · Your role: Data Scientist on the engagement
4-Week Engagement · Python · pandas · scikit-learn · Streamlit

PROGRAM Zidio Internship	DOMAIN Data Science & Analytics	DURATION 4 Weeks (1 Month)
ENGAGEMENT TYPE client project	ROLE Data Scientist & Analytics	PROJECT WEIGHT 50 / 100 Marks

Document Control

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How to read this brief

This is written as a client engagement, not a coursework assignment. Treat NorthBay Living as a real customer: they have a business problem, a budget of your time, and stakeholders who will judge the work on whether it helps them decide what to stock.

Sections 01–04 set the context and what you must deliver. Section 09 lists each deliverable with its acceptance criteria. Sections 11–13 are how you are graded and what you hand in. Read those before you start.

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01 Engagement Overview

Project FORESIGHT is a four-week data science engagement. You are not completing a coursework assignment — you are the data scientist Zidio has placed on a client account, and you are expected to behave like one: understand the client’s business, deliver analysis they can act on, and present it so a non-technical decision-maker trusts it.

The client, NorthBay Living, is a mid-size direct-to-consumer home & lifestyle brand. They sell online, they hold inventory, and right now they plan that inventory on gut feel and spreadsheets. They are losing money in two directions at once: best-sellers run out (lost sales) and slow movers pile up (cash locked in stock that later gets marked down). Your job is to turn their own data into a demand forecast and an early-warning system that tells the planning team what to reorder, what to clear, and what to leave alone.

1.1 What you will deliver

By the end of the engagement you will hand the client a working forecasting and risk system, a planning dashboard, a deployed scoring service, and an executive readout. The full deliverable list and acceptance criteria are in Sections 04 and 09.

02 Client Background & The Brief

2.1 The client

NorthBay Living	
Business	Direct-to-consumer home & lifestyle brand (furnishings, décor, small appliances).
Scale	~200 active SKUs across several categories; online-only; ships from one warehouse.
Data maturity	Sales and inventory live in exports and spreadsheets; no forecasting in place today.
Who you work with	Head of Operations (your main point of contact), a merchandiser, and the Finance lead.

2.2 The brief, in the client’s words

From: Head of Operations, NorthBay Living

“Every month we stock out of things people want and sit on things they don’t. We’re guessing how much to order and when. I need something that tells me, for each product, how much we’ll likely sell over the next few weeks, which products are about to run out, and which ones are overstocked so we can clear them. And it has to be something my team can actually look at — not a notebook only a data scientist can read.”

That brief contains four asks: a demand forecast, a stockout early-warning, an overstock flag, and a usable interface for a non-technical team. Everything in this engagement maps back to those four asks.

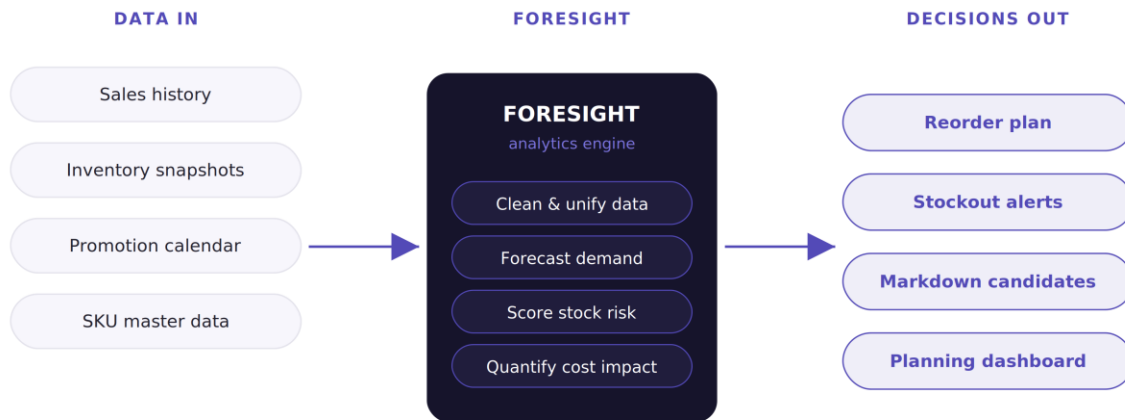


Figure 1. How FORESIGHT turns NorthBay's raw data into stocking decisions the team can act on.

2.3 Why it matters to the business

Stockouts cost NorthBay sales they can never recover and frustrate customers. Overstock locks up cash and ends in margin-eroding markdowns. Even a modest improvement in stocking accuracy frees working capital and protects revenue, which is why this work has executive attention.

03 Objectives & Success Metrics

3.1 Engagement objectives

1. Produce a weekly, SKU-level demand forecast for a defined horizon that measurably beats a naive baseline.
2. Flag stockout and overstock risk for every SKU over the forecast horizon, with a clear recommended action.
3. Quantify the business impact in rupees (sales at risk from stockouts, capital locked in overstock).
4. Hand the operations team a dashboard they can use without you, plus a deployed service that returns a forecast and risk for any SKU.
5. Present findings to stakeholders in language a non-technical decision-maker trusts and can act on.

3.2 Success metrics (how the client judges success)

These are the KPIs the engagement is held to. They are written as targets, the way a client would state them.

Success metric	Target	Why the client cares
Forecast accuracy (WAPE)	Beats seasonal-naive baseline by a clear margin on backtest	An unreliable forecast is worse than none — it must earn trust.
Risk precision	Stockout / overstock flags are right often enough to act on	False alarms waste the team's time; misses cost money.
Quantified impact	Rupee figure for sales-at-risk and locked capital	Turns analysis into a business case leadership understands.
Usability	Ops team can answer "what do I reorder?" unaided	If they can't use it, it won't change decisions.
Reproducibility	Pipeline re-runs end-to-end from raw data	The client must be able to refresh it next month.

3.3 Learning outcomes

Delivering the above proves, with evidence in your repository, that you can:

- Build a reproducible data pipeline from messy real-world extracts to an analysis-ready dataset.
- Run rigorous EDA and translate patterns into business insight, not just charts.
- Engineer features and build, backtest, and honestly evaluate a forecasting model against a baseline.
- Turn predictions into decisions through a transparent risk-scoring rule.
- Ship analytics as a product: a dashboard, a deployed service, and a stakeholder readout.

04 Scope & Deliverables

Scope is fixed so the engagement is finishable in four weeks and graded fairly. Deliver everything in scope to a high standard before considering anything beyond it.

4.1 Deliverables (what the client receives)

#	Deliverable	What it is
D1	Data pipeline	Reproducible ingestion + cleaning producing an analysis-ready dataset.
D2	Data-quality & EDA insight memo	Findings on patterns, seasonality, top movers, dead stock, and data issues.
D3	Demand forecast model	Weekly SKU-level forecast, backtested, beating a naive baseline.
D4	Risk scoring	Stockout / overstock classification with recommended actions per SKU.
D5	Planning dashboard	Interactive dashboard for the ops team (forecast, risk, reorder view).
D6	Deployed scoring service	A hosted app/API returning forecast + risk for a SKU or batch.

#	Deliverable	What it is
D7	Executive readout	A stakeholder deck/memo: findings, rupee impact, and recommendations.

4.2 In scope

- Weekly demand forecasting at SKU level over a defined horizon (e.g., 6–8 weeks).
- Stockout and overstock risk scoring driven by the forecast and inventory position.
- One reproducible pipeline, a dashboard, a deployed scoring service, and an executive readout.

4.3 Out of scope (do not build)

Explicitly excluded — building these instead of the required scope will lower your score.

- Live integrations with the client’s real systems (you work from the provided extracts).
- Price optimization, supplier selection, or full inventory-optimization solvers.
- Real-time / streaming pipelines and big-data infrastructure.
- Automated purchase-order placement or any action taken on the client’s behalf.

Definition of done

A deliverable is “done” only when it meets its acceptance criteria in Section 09, runs reproducibly from the raw data, and a stakeholder could use it without you in the room. “It works on my machine” is not done.

05 Data Provided & Data Dictionary

You receive four simulated extracts modelled on what a real D2C brand would have. They are deliberately imperfect — expect missing values, the odd duplicate, and inconsistent labels — because cleaning them is part of the work. The schema below shows how they relate; full column-level detail is in Appendix A.

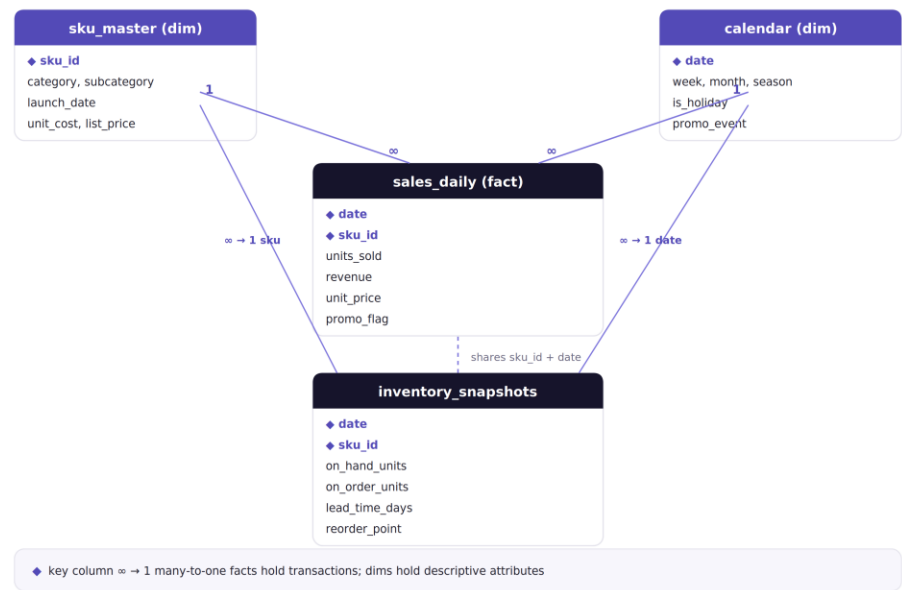


Figure 2. The provided datasets form a simple star schema: one transactional fact table joined to descriptive dimensions.

5.1 The four tables

Table	Grain & contents
sales_daily	One row per SKU per day: units sold, revenue, price, promo flag. The fact table.
sku_master	One row per SKU: category, subcategory, launch date, unit cost, list price.
calendar	One row per date: week, month, season, holiday flag, promotion events.
inventory_snapshots	Periodic stock position per SKU: on-hand, on-order, lead time, reorder point.

Data is provided — generating it is not your job

A synthetic dataset (and a generator, if you want more) accompanies this brief. Do not spend the engagement fabricating data. Spend it cleaning, understanding, and modelling the data you are given, exactly as you would with a real client extract.

06 Solution Architecture

FORESIGHT is a straightforward analytics stack: raw extracts flow through a reproducible pipeline into models, and the models’ outputs are served to stakeholders through a dashboard and a scoring service. Keep the layers separate and documented so the client can re-run and trust the whole thing.

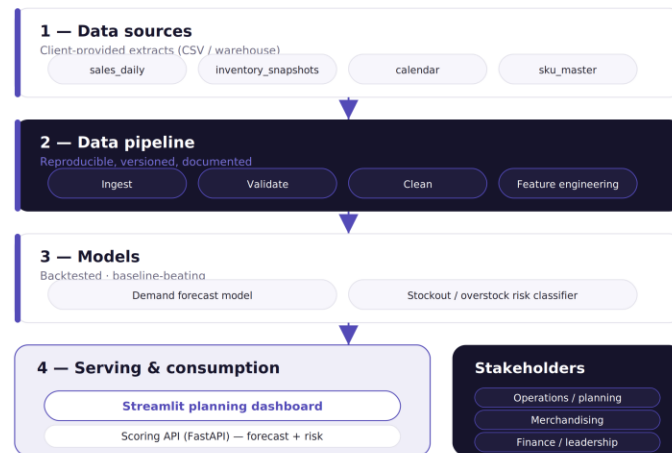


Figure 3. Solution architecture — from client extracts through a reproducible pipeline to models and stakeholder-facing tools.

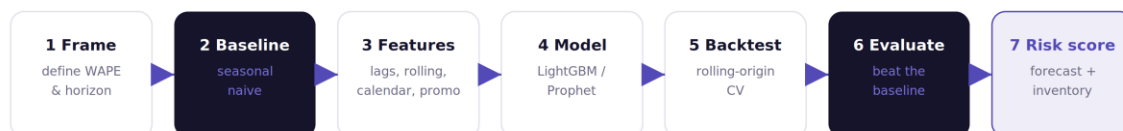
6.1 Principles

- **Reproducible**: the whole pipeline re-runs from raw data with one command; no manual spreadsheet steps.
- **Honest**: every model is compared against a baseline on a fair backtest before it is trusted.
- **Usable**: outputs reach stakeholders as a dashboard and a service, not only as notebooks.
- **Documented**: assumptions, data issues, and limitations are written down, not left implicit.

07 Methodology & Modelling Approach

Follow a disciplined workflow rather than jumping to a complex model. The order below is deliberate: you define how you will measure success, build a baseline, and only then try to beat it. This is how professional forecasting work avoids fooling itself.

MODELLING WORKFLOW — BASELINE FIRST, THEN BEAT IT



The baseline is the bar. If a complex model can't beat seasonal-naive on backtest, ship the baseline and say so.
Honest accuracy beats an impressive-looking model that doesn't generalise.

Figure 4. The modelling workflow: frame the metric, build a baseline, then earn the right to use a more complex model.

7.1 Step detail

Stage	What you do
Frame & metric	Fix the forecast horizon and the accuracy metric (WAPE, with bias as a secondary check) before modelling.
Baseline	Build a seasonal-naive forecast. This is the bar every later model must clear.
Features	Engineer lag features, rolling statistics, calendar/seasonality, and promotion signals.
Model	Train a forecasting model (e.g., gradient-boosted trees or a classical time-series method).
Backtest	Evaluate with rolling-origin cross-validation — never a single random split for time series.
Evaluate & select	Compare to the baseline on backtest. If it doesn't win, ship the baseline and say why.
Risk score	Combine the forecast with inventory position to score stockout and overstock risk.

The non-negotiable rule of this engagement

Beat the baseline, honestly. Forecasting is uniquely easy to fake by leaking future information or testing on the training period. Use rolling-origin backtesting, never let future data touch a feature, and report WAPE against the baseline in your readme. A model that can't beat seasonal-naive is not a failure to hide — it is a finding to report.

7.2 Illustrative output

Your forecast output should look like the figure below: actual history, the baseline, and your model's forecast with an uncertainty interval over the horizon. The interval matters — the client needs to know how confident the forecast is, not just a single number.

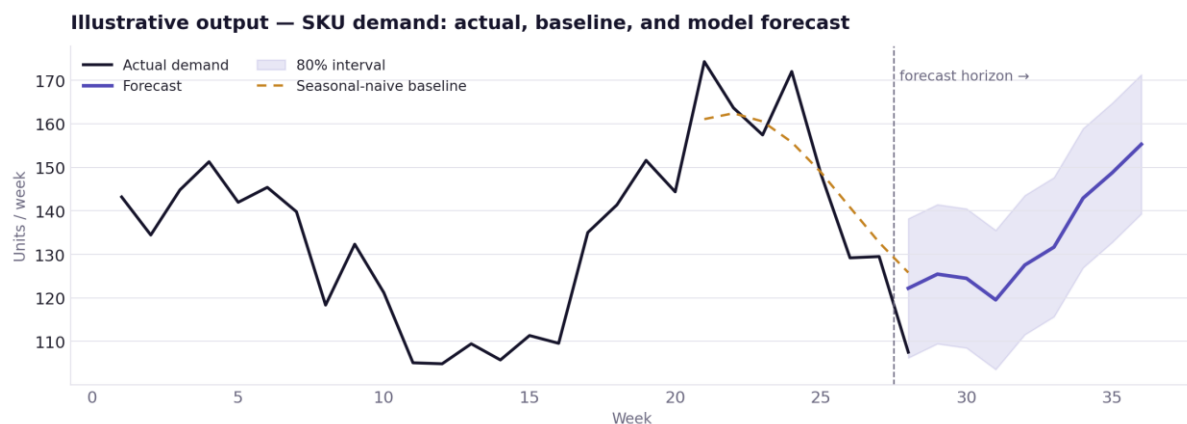


Figure 5. Illustrative forecast output — actual demand, the seasonal-naive baseline, and the model forecast with an 80% interval.

08 Risk Scoring & Decisioning

A forecast on its own does not tell the ops team what to do. The risk layer converts the forecast plus the current inventory position into a decision for every SKU. This is the part the client actually acts on, so it must be transparent and explainable.

8.1 How risk is scored

- Stockout risk: compare forecast demand over the lead time against on-hand plus on-order stock. If projected stock falls below a safety level, the SKU is at risk of stocking out.
- Overstock risk: compare on-hand stock against forecast demand over a forward window. If you are holding far more than you will sell, the SKU is overstocked.
- Each SKU gets a risk level and a recommended action; the rupee value at stake is attached so the team can prioritise.

8.2 The decisioning view

Plot every SKU on a stockout-versus-overstock grid. The four quadrants map directly to actions, so the ops team can triage hundreds of SKUs at a glance instead of reading a table row by row.

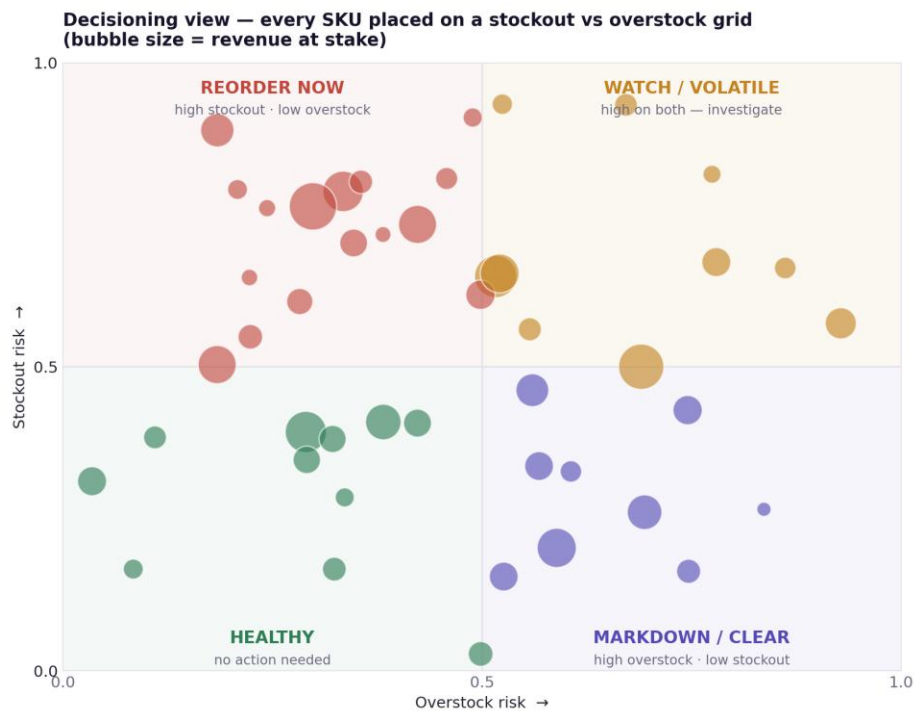


Figure 6. Decisioning view — each SKU placed on a stockout vs overstock grid, sized by revenue at stake.

Quadrant	Meaning	Recommended action
Reorder now	High stockout, low overstock	Raise a replenishment order before stock runs out.
Markdown / clear	High overstock, low stockout	Promote or discount to free up capital.
Watch / volatile	High on both	Investigate — demand is erratic; review manually.
Healthy	Low on both	No action needed; leave as is.

09 Deliverable Specifications

Each deliverable has acceptance criteria — the client's definition of done. Treat them as your checklist; mentors grade against them.

D1 Data pipeline

Acceptance criteria

1. Ingests all four extracts and produces one analysis-ready dataset.
2. Cleaning steps (missing values, duplicates, type fixes) are coded, not manual.
3. Re-runs end-to-end from raw files with a single command.
4. Key cleaning decisions are documented with their rationale.

D2 Data-quality & EDA insight memo

Acceptance criteria

1. Reports data-quality issues found and how they were handled.
2. Shows demand patterns: seasonality, trend, top movers, and dead stock.
3. States at least three business-relevant insights in plain language.
4. Charts are labelled and readable by a non-technical reader.

D3 Demand forecast model

Acceptance criteria

1. Produces a weekly SKU-level forecast over the defined horizon.
2. Includes a seasonal-naive baseline for comparison.
3. Backtested with rolling-origin CV; reports WAPE vs the baseline.
4. No data leakage — future information never enters a feature.

D4 Risk scoring

Acceptance criteria

1. Scores stockout and overstock risk for every SKU over the horizon.
2. Attaches a recommended action and the rupee value at stake.
3. Logic is transparent and explainable, not a black box.
4. Output reconciles with the decisioning grid in Section 08.

D5 Planning dashboard

Acceptance criteria

1. Filterable by category and SKU; shows forecast vs actual and risk flags.
2. Surfaces a prioritised reorder / markdown list.
3. Loads on seeded data and is usable by a non-technical person.
4. Has clear empty and loading states.

D6 Deployed scoring service

Acceptance criteria

1. Hosted and reachable via a public URL.
2. Returns forecast + risk for a given SKU (or a batch).
3. Documented inputs and outputs.
4. Handles bad input gracefully rather than crashing.

D7 Executive readout

Acceptance criteria

1. A 6–10 slide deck (or memo) aimed at the Head of Operations and Finance.
2. Leads with the rupee impact and the recommended actions.
3. Explains accuracy and limitations honestly.
4. Contains no unexplained jargon — a non-technical reader can follow it.

10 Engagement Timeline — 4-Week Plan

The engagement runs four weeks (twenty working days). Each week ends in a client checkpoint where you demo progress against that week's deliverable. Work the weeks in order — the methodology depends on it.

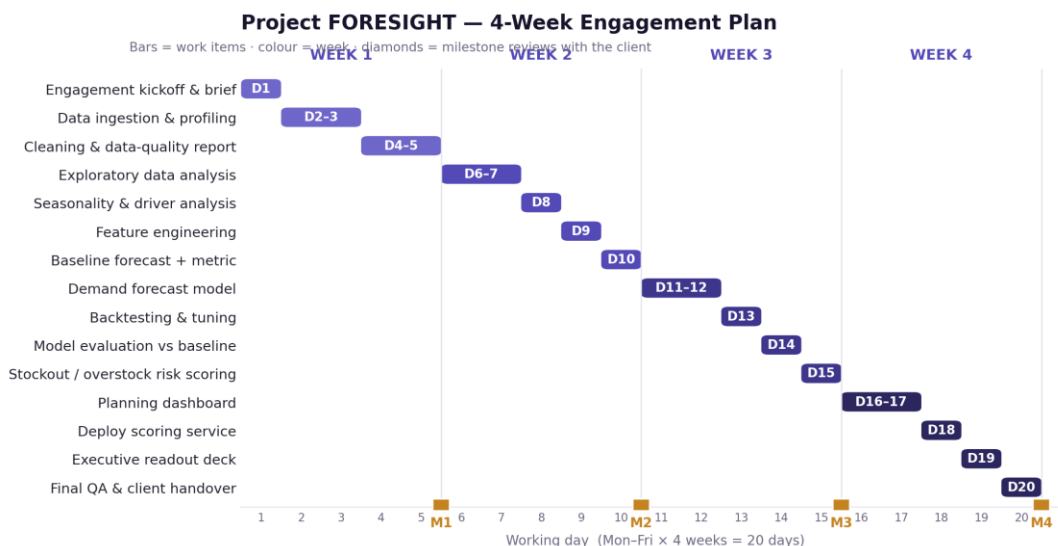


Figure 7. The four-week engagement plan, with client checkpoints at the end of each week (M1–M4).

WEEK 1	
1	
Data Foundation	
Client checkpoint deliverable: A reproducible pipeline and a data-quality report on cleaned data.	
Day 1	Engagement kickoff: read the brief, restate the problem and success metrics in your own words, set up the repo and environment.

Day 2–3	Ingest all four extracts; profile them; document structure, grain, and obvious issues.
Day 4–5	Clean and unify into an analysis-ready dataset; write the data-quality report. Checkpoint M1.

WEEK

2

EDA & Baseline

Client checkpoint deliverable: An EDA insight memo and a working seasonal-naive baseline with a chosen metric.

Day 6–7	Exploratory analysis: demand distributions, top movers, dead stock, correlations.
Day 8	Seasonality and driver analysis (calendar, promotions, category effects).
Day 9	Feature engineering: lags, rolling statistics, calendar and promo features.
Day 10	Build the seasonal-naive baseline; fix WAPE as the metric. Checkpoint M2.

WEEK

3

Modelling & Risk

Client checkpoint deliverable: A backtested forecast that beats the baseline, plus stockout / overstock risk scoring.

Day 11–12	Train the demand forecast model on engineered features.
Day 13	Rolling-origin backtesting and tuning; guard against leakage.
Day 14	Evaluate against the baseline; select the model (or justify keeping the baseline).
Day 15	Build stockout / overstock risk scoring and the decisioning grid. Checkpoint M3.

WEEK

4

Productize & Present

Client checkpoint deliverable: Dashboard, deployed scoring service, and an executive readout.

Day 16–17	Build the planning dashboard (forecast, risk, reorder/markdown list).
Day 18	Deploy the scoring service; smoke-test the public URL.
Day 19	Quantify rupee impact; build the executive readout deck/memo.
Day 20	Final QA against acceptance criteria; record demo; client handover. Checkpoint M4.

If you fall behind

Protect the forecast, the risk scoring, and the deployment (Weeks 3–4) — they carry the most marks and are what the client acts on. Trim the depth of EDA or the polish of the dashboard before you trim core modelling. Tell your mentor early if you are blocked.

11 Milestones & Acceptance

Four milestones gate the engagement, each ending in a client checkpoint where you demo the deliverable. Hitting them on time is part of delivering like a professional.

Milestone	End of	Deliverable	Project pts	Status
M1 — Data foundation	Week 1	Reproducible pipeline + data-quality report.	10	☐
M2 — EDA & baseline	Week 2	EDA insight memo + seasonal-naive baseline & metric.	10	☐
M3 — Modelling & risk	Week 3	Backtested forecast beating baseline + risk scoring.	20	☐
M4 — Productize	Week 4	Dashboard + deployed service + executive readout.	10	☐

Why M3 carries the most marks

Modelling and risk scoring are the data-science core of this engagement and the hardest to fake. A clean pipeline and a pretty dashboard around a forecast that doesn't beat the baseline is not a passing project. Spend your best effort in Week 3.

11.1 Optional stretch goals

Only after the full scope is excellent. These strengthen a strong submission; they never substitute for core work.

- Prediction intervals calibrated and reported, not just point forecasts.
- A simple what-if control in the dashboard (e.g., adjust lead time, see risk change).
- Feature-importance or explainability for the forecast drivers.
- A short model-monitoring plan: how NorthBay would detect the forecast degrading.

12 Evaluation & Scoring

The engagement is scored within Zidio's standard 100-point rubric. This brief governs the Project (50) and Submission (20) components; Attendance (20) and Test (10) are assessed by the program separately.

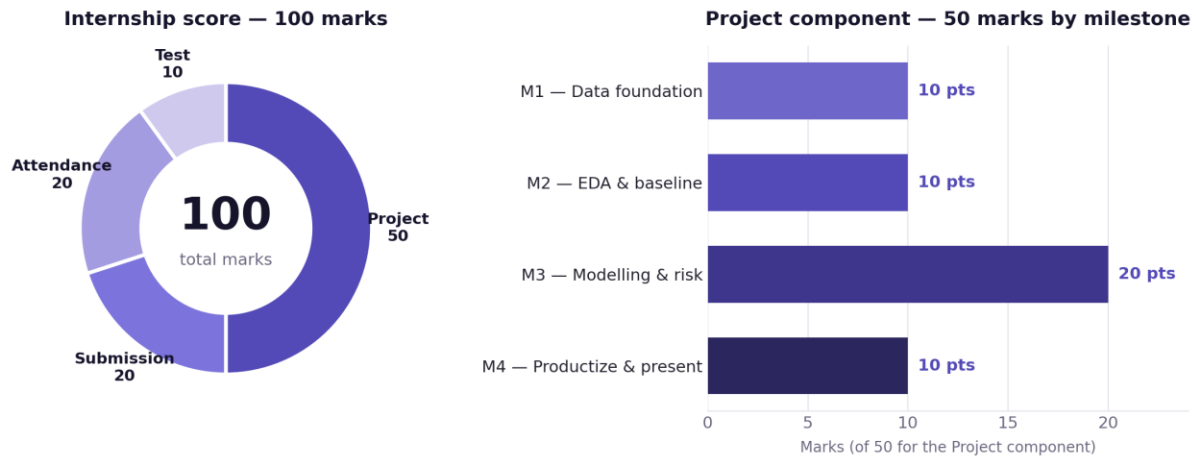


Figure 8. Mark distribution across the 100-point rubric, and the 50-point project broken down by milestone.

12.1 Project breakdown (50 marks)

Milestone	Marks	What earns full marks
M1 Data foundation	10	A genuinely reproducible pipeline and an honest data-quality report — issues found and handled, not hidden.
M2 EDA & baseline	10	Insight (not just charts), sound features, and a correct baseline with the metric defined.
M3 Modelling & risk	20	A leakage-free, backtested forecast that beats the baseline, plus transparent risk scoring tied to decisions.
M4 Productize	10	A usable dashboard, a working deployed service, and a readout a non-technical stakeholder trusts.

12.2 Submission breakdown (20 marks)

Criterion	Marks	What earns full marks
Repository & notebooks	5	Clean, reproducible, well-documented; clear README with setup and results.
Dashboard & deployment	5	A live, usable dashboard and a reachable scoring service.
Executive readout	5	Clear, honest, decision-focused; rupee impact up front.
Demo video	5	A 3–5 minute walkthrough of the analysis and the tools, end to end.

Quality bar — what separates a top score

Average submission: a model with a good-looking accuracy number, but no baseline comparison, shaky validation, and outputs only a data scientist could read.

Top submission: an honestly-validated forecast that beats the baseline, risk scoring that drives clear actions, and a readout that would actually change what NorthBay orders next month.

13 Submission Requirements

Submit all of the following by the cohort deadline. A missing item is a missing mark.

1. Git repository (public, or private with mentor access) containing the pipeline, notebooks, and model code.
2. Live dashboard URL and live scoring-service URL.
3. A README with: the problem, the data, setup/run steps, the backtest result (WAPE vs baseline), and key assumptions.
4. The executive readout (PDF/slides) and the data-quality & EDA memo.
5. A 3–5 minute demo video (unlisted link) walking through the analysis and the tools.
6. The completed cohort submission form with all links.

Reproducibility check

A grader should be able to clone your repo, follow the README, and re-run your pipeline to the same headline numbers. If it only runs on your laptop, treat that as a defect to fix before submitting.

14 Ways of Working & Tooling

14.1 Cadence

Ritual	Purpose
Daily check-in (async)	A short written update: what you did, what's next, any blockers.
Weekly client checkpoint	Demo the week's deliverable to your mentor (the "client"); get direction. Mandatory.
Mentor office hours	Technical blockers, method questions, and review.

14.2 Recommended toolset

Area	Tools
Language & analysis	Python, pandas, numpy, Jupyter for exploration.
Modelling	scikit-learn, plus a forecasting approach (statsmodels / Prophet / LightGBM).
Visualisation	matplotlib / plotly; a clean, labelled style.
Dashboard	Streamlit (fast to build and deploy).
Serving	FastAPI for the scoring service (optional if the dashboard exposes scoring).
Deployment	Streamlit Community Cloud / Hugging Face Spaces / Render.
Version control	Git; meaningful commits; no data dumps or secrets committed.

14.3 Repository structure (suggested)

```
foresight/
  data/          # raw + processed (gitignored if large)
  notebooks/     # 01_eda.ipynb, 02_baseline.ipynb, 03_model.ipynb
  src/
    pipeline.py  # ingest + clean + features (reproducible)
    forecast.py  # model train / predict / backtest
    risk.py      # stockout / overstock scoring
  app/           # Streamlit dashboard
  service/       # FastAPI scoring endpoint
  reports/       # EDA memo, executive readout
  README.md
  requirements.txt
```

15 Stakeholders & RACI

Knowing who you serve and who decides is part of working like a consultant. On this engagement the roles are simple.

Stakeholder	Interest
Head of Operations	Primary client. Wants reliable reorder/markdown guidance the team can use.
Merchandise	Cares about which products to push, clear, or discontinue.
Finance lead	Cares about cash freed and revenue protected — the rupee impact.
You (Data Scientist)	Owens the analysis, the tools, and the readout.
Zidio mentor	Acts as engagement lead / client proxy at checkpoints.

15.1 RACI (who does what)

Activity	Intern	Mentor	Ops	Finance
Build pipeline & model	R / A	C	I	I
Define success metrics	R	A	C	C
Risk scoring & actions	R / A	C	C	I
Dashboard & deployment	R / A	C	I	I
Executive readout	R	C	I	A

R = Responsible · A = Accountable · C = Consulted · I = Informed.

16 Risks, Assumptions & Data Governance

16.1 Assumptions

- The provided extracts are representative of NorthBay's real sales and inventory behaviour.
- History is long enough to learn weekly seasonality for most SKUs.

- Lead times and reorder points in the inventory data are broadly accurate.

16.2 Risks & mitigations

Risk	Mitigation
Sparse history for new SKUs	Fall back to category-level patterns; flag low-confidence forecasts.
Promotions distort demand	Include promo features; report forecasts with and without promo effects.
Overfitting / leakage	Rolling-origin backtest; strict separation of past and future in features.
Dashboard nobody uses	Design with the ops persona in mind; keep it to the few questions they ask.

16.3 Data governance

- Treat the (simulated) client data as confidential; do not post it publicly or commit large raw dumps.
- Keep the pipeline reproducible and the random seeds fixed so results can be re-created.
- Document every assumption and cleaning decision — the client must be able to audit the work.

17 Originality & Professional Conduct

- You may use AI assistants, libraries, and documentation — that is normal practice. You must understand and be able to explain every result you present.
- You may reference tutorials, but the analysis, modelling choices, and risk logic must be your own work on this dataset.
- Never fabricate results or hide a poor backtest. Reporting that the baseline won is honest and acceptable; faking that the model won is misconduct.
- At the final review, mentors may ask you to explain or modify any part of your pipeline or model live.

18 Appendix A — Data Dictionary

sales_daily

Column	Description
date	Calendar date of the sales record.
sku_id	Product identifier (joins to sku_master).
units_sold	Units sold that day.
revenue	Revenue for that SKU-day.
unit_price	Selling price that day.

Column	Description
promo_flag	1 if the SKU was on promotion that day, else 0.

sku_master

Column	Description
sku_id	Product identifier (primary key).
category / subcategory	Product classification.
launch_date	When the SKU was first sold.
unit_cost / list_price	Cost and list price for margin and impact.

calendar

Column	Description
date	Calendar date (primary key).
week / month / season	Time attributes for seasonality.
is_holiday	Holiday indicator.
promo_event	Named promotional event, if any.

inventory_snapshots

Column	Description
date / sku_id	Snapshot date and product.
on_hand_units	Stock physically available.
on_order_units	Stock ordered, not yet received.
lead_time_days	Replenishment lead time.
reorder_point	Stock level that should trigger a reorder.

19 Appendix B — Metric Definitions

Metric	Definition & why it is used
WAPE	Weighted Absolute Percentage Error — total absolute error divided by total actual demand. Primary accuracy metric; robust to low-volume SKUs where MAPE explodes.
MAPE	Mean Absolute Percentage Error — reported as a secondary, intuitive figure, but unreliable when demand is near zero.
Bias	Mean forecast error (signed) — checks whether the model systematically over- or under-forecasts.

Metric	Definition & why it is used
Seasonal-naive	Baseline that predicts demand equal to the same period last season; the bar every model must beat.
Rolling-origin CV	Backtesting that repeatedly trains on the past and tests on the next period, mimicking real forecasting.
Service level	Probability of not stocking out over the lead time; underpins the stockout-risk threshold.

20 Appendix C — Resources

Topic	Link
pandas	https://pandas.pydata.org/docs
scikit-learn	https://scikit-learn.org/stable
statsmodels	https://www.statsmodels.org/stable
Prophet	https://facebook.github.io/prophet
LightGBM	https://lightgbm.readthedocs.io
Streamlit	https://docs.streamlit.io
FastAPI	https://fastapi.tiangolo.com

21 Appendix D — Glossary

Term	Meaning
SKU	Stock-Keeping Unit — a single sellable product variant.
Stockout	Running out of stock so demand cannot be met — lost sales.
Overstock	Holding far more stock than will sell — locked capital and markdown risk.
Lead time	Time between ordering stock and it arriving and being sellable.
Reorder point	Stock level at which a replenishment order should be placed.
Backtest	Testing a forecast on history it did not train on, to estimate real accuracy.
Baseline	A simple reference forecast a real model must beat to be worth using.
Working capital	Cash tied up in operations — including stock sitting in the warehouse.